

LOSS-INFORMED TRANSFORMER TRAINING WITH LEARNED RATIONAL ACTIVATIONS

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ABSTRACT

Learning an activation changes how weight updates propagate through a network. We use this evolving response to improve Transformer pretraining through loss-informed coordination of matrix updates. GRAIN (Groupwise Rational Activation with Internal Normalization) learns rational response shapes in normalized feature coordinates and restores the group scale at its output. TILLER (Tangent-Informed Loss Ledger for Equal-budget Reweighting) uses these learned responses to orient matrix directions and measure their loss contributions at shared activation positions. A compact loss ledger retains a history of these responses, capturing how directions reinforce or offset one another at common positions. Combining this information with current gradients, TILLER jointly allocates update strength across groups and layers while preserving the combined direction norm of the feed-forward updates at each step. In language-model pretraining, GRAIN with TILLER achieves the lowest mean validation loss among the evaluated adaptive and matrix-based optimizer baselines across every tested corpus and model size at matched training-token budgets.

1 INTRODUCTION

Learning an activation function gives a Transformer control over both its feature representation and the response of those features to weight updates. Fixed nonlinearities such as GELU and the SiLU gate in SwiGLU provide effective transformations (Hendrycks & Gimpel, 2016; Shazeer, 2020). Rational activations make the response shape trainable (Molina et al., 2020), with approximation advantages that motivate their use as expressive neural representations (Boullé et al., 2020; Tang & Townsend, 2026). Group-shared rational functions bring this flexibility to Transformer feed-forward blocks (Yang & Wang, 2025). The changing response also supplies information for optimizing the matrices around it: learning the nonlinearity changes how an update propagates through the network.

The incoming and outgoing projections expose two complementary paths. An incoming update changes features through the activation Jacobian; an outgoing update recombines the activated features. Relating these changes to the downstream loss gives a functional description of each matrix direction. The interaction between directions is equally important. Two groups can have the same average descent signal while their loss responses reinforce or offset one another at shared activation positions. Separate descent scores leave these relationships unresolved. Measuring the responses together makes them available for choosing the strengths of simultaneous updates.

We introduce GRAIN, Groupwise Rational Activation with Internal Normalization, as the learned representation for this construction. Each feature group evaluates its own rational function on RMS-normalized coordinates and multiplies the result by the same normalization scale (Zhang & Sennrich, 2019). The learned coefficients control the response to relative feature magnitudes, while the output retains a dependence on input amplitude. Its Jacobian describes both the local rational slope and the effect of changing the group scale. The features and their derivatives provide the two response paths used to guide optimization.

Matrix-aware optimization uses matrix structure and activation sensitivity to shape updates (Martens & Grosse, 2015; Vyas et al., 2025; Liu et al., 2025). TILLER, Tangent-Informed Loss Ledger for Equal-budget Reweighting, uses the learned responses to construct matrix directions and coordinate their joint effect on the loss. It uses the response geometry to rotate an orthogonalized momentum direction along a factorized adaptive tangent, with orientation selected by the gradient (Shazeer &

Stern, 2018). It then records the selected feed-forward directions’ loss responses at shared positions in a compact temporal ledger. Together with the current gradients, this history defines a joint model for allocating update strength across groups and layers. The allocation preserves the combined norm of that step’s feed-forward directions, so the ledger changes where the update acts according to the responses it produces.

This construction gives two training methods. Full TILLER continually adapts directions and their joint allocation throughout training. The staged method applies TILLER for an initial interval, then continues from the learned model with a chosen optimizer. GRAIN remains trainable in both methods. Full TILLER maintains loss-response coordination along the complete trajectory; staged training combines a coordinated training stage with the update rule and computational cost of its continuation optimizer.

Across five pretraining corpora and both evaluated model sizes, GRAIN with full TILLER achieves the lowest mean validation loss among the tested activation–optimizer configurations at matched token budgets. The comparisons include AdamW, Muon, SOAP, and factorized adaptive optimizers. At the larger model size, the evaluated staged configuration also achieves lower mean validation loss than every baseline, while using less total training time than full TILLER on every corpus.

[TODO: Experiment section]

2 RELATED WORK

Padé Activation Units learn rational curves jointly with network weights (Molina et al., 2020). Kolmogorov–Arnold Transformer (KAT) implements group-shared rational functions followed by linear maps; its reported configuration shares numerator coefficients across groups (Yang & Wang, 2025). GRAIN evaluates each group’s rational function in RMS-normalized coordinates (Zhang & Sennrich, 2019) and restores the group scale at its output. It retains group sharing while learning both numerator and denominator coefficients independently for each group.

Approximation theory establishes parameter-efficiency separations between rational and ReLU networks (Boullé et al., 2020). *Rational Neural Networks have Expressivity Advantages*, by Maosen Tang and Alex Townsend, extends these comparisons to bounded-parameter smooth-activation networks and analyzes coefficient redundancy under learned affine normalization (Tang & Townsend, 2026). GRAIN uses an internal normalization–rescaling map with no additional learned affine transform.

Muon orthogonalizes momentum (Jordan et al., 2024), while Adafactor estimates gradient second moments from row and column statistics (Shazeer & Stern, 2018). TILLER combines these direction constructions using the learned activation response. Loss-weighted features and Jacobians determine the extent of a rotation from the polar direction along an adaptive tangent, with its orientation selected by the current gradient.

Joint update models differ in their search directions and the information used to compare them. K-FAC constructs parameter-space Fisher approximations from activation and backpropagated-derivative moments, including cross-layer blocks (Martens & Grosse, 2015). Newton–Muon derives input-second-moment preconditioning from a matrix quadratic surrogate (Du & Su, 2026). Krylov Subspace Descent optimizes the nonlinear objective over a curvature-generated Krylov subspace (Vinyals & Povey, 2012). TILLER places its quadratic model on coefficients multiplying selected pairs of incoming and outgoing group directions. Current gradients supply the linear term. A discounted history of shared-position response probes, augmented between probes by current-gradient information, supplies the quadratic penalty. Its cross terms couple coefficients across groups and layers; the allocation preserves the combined feed-forward direction norm before weight decay.

Optimizer switching combines update rules along a training trajectory. SWATS selects an Adam-to-SGD transition using a projection-based trigger (Keskar & Socher, 2017). Our staged method applies TILLER until a chosen step, then continues with a selected optimizer while retaining the learned parameters and trainable GRAIN activations.

3 METHOD

GRAIN learns the nonlinear response of a Transformer feed-forward block. TILLER uses that response to orient matrix updates and coordinate their strengths across groups and layers. This construction supports both full-horizon optimization and staged training.

3.1 GRAIN

GRAIN (Groupwise Rational Activation with Internal Normalization) assigns a learned rational curve to each group of intermediate features (Molina et al., 2020; Yang & Wang, 2025). For an input $x \in \mathbb{R}^d$, partition the intermediate features into G groups of width w , with incoming and outgoing projections $A_g \in \mathbb{R}^{w \times d}$ and $B_g \in \mathbb{R}^{d \times w}$. Writing $z_g = A_g x$, the block computes

$$\rho_g = \sqrt{\|z_g\|_2^2/w + \epsilon_{\text{act}}}, \quad h_g = \rho_g \phi_g(z_g/\rho_g), \quad y = \sum_{g=1}^G B_g h_g. \quad (1)$$

The RMS scale ρ_g depends on the current token and group (Zhang & Sennrich, 2019). Dividing by it presents relative feature magnitudes to the learned curve; multiplying by it carries the group amplitude into the output. The positive stabilizer ϵ_{act} keeps the map defined at zero.

The scalar curve is applied coordinatewise:

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

Its denominator is at least one. Each group learns its own numerator and denominator coefficients, shared across features and token positions, from a common SiLU approximation. The coefficients and projection weights are trained jointly. Because the normalization scale also changes with the input, the group Jacobian $J_g = \partial h_g / \partial z_g$ contains both the coordinatewise rational slopes and coupling through the shared scale. An incoming direction U_g acts through $B_g J_g U_g x$, whereas an outgoing direction V_g acts through $V_g h_g$. These are the two response paths used by TILLER. Appendix A derives their evaluation.

3.2 TILLER

TILLER (Tangent-Informed Loss Ledger for Equal-budget Reweighting) first selects each matrix direction, then allocates a common feed-forward update norm. For L layers, index the $C = LG$ layer-group pairs by $i = (\ell, g)$ and the two matrix roles by $r \in \{A, B\}$. Write g_i^r for the accumulated, averaged, and clipped gradient. For each matrix, a group-calibrated polar parent p_i^r and a factorized adaptive direction a_i^r transform the same momentum (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). The polar map rebalances singular values, while the adaptive direction incorporates row and column gradient scales. The component of a_i^r orthogonal to p_i^r supplies a tangent for rotating the update.

The polar parent provides a reference direction. TILLER admits a larger adaptive tangent when the learned response is both broad and different from its initial geometry. Participation $\pi_i^r \in [0, 1]$, weighted by local loss sensitivity, measures energy spread across Jacobian modes for the incoming path and feature coordinates for the outgoing path. Congruence $\kappa_i^r \in [0, 1]$ is the Frobenius cosine between collections of current kernels $J_i J_i^\top$ or $h_i h_i^\top$ and their initial-coefficient counterparts, with the same weights and current inputs. Suppressing the role index,

$$\begin{aligned} a_i^\perp &= a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, & e_i &= \pi_i (1 - \kappa_i^2), \\ \tilde{q}_i &= \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \end{aligned} \quad (3)$$

The sign σ_i makes the tangent’s gradient contraction nonnegative, and orthogonality preserves the parent norm. Aligned kernels retain the parent. A finite, nondegenerate candidate is selected when both it and its parent have positive gradient contractions; otherwise the parent is retained. A matrix-size multiplier gives the selected direction q_i^r .

One signed coefficient c_i scales the paired directions (q_i^A, q_i^B) . To compare their simultaneous effects, TILLER measures responses at K example–token positions shared across layers. Let $x_{k\ell}$ be the block input and $\xi_{k\ell}$ the derivative of the summed minibatch loss with respect to the block output, multiplied by the realized gradient-clipping factor. The response matrix contains

$$S_{t,ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle. \quad (4)$$

Thus $S_t c$ combines the groups’ contributions through common activation positions. The cross terms in $\|S_t c\|_2^2 / K$ distinguish reinforcement and cancellation that individual descent scores leave unresolved.

The loss ledger stores a discounted second moment of these joint responses. Periodic probes measure positional variation; between probes, rows aligned with current gradient contractions update the history at the latest measured response scale. A Robust Frequent Directions factor and an isotropic remainder compress this history (Ghashami et al., 2016; Luo et al., 2019). Fresh rows enter the current metric G_t before compression.

TILLER allocates the update norm by balancing current descent against this response history. The full clipped gradients give $s_{t,i} = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$. With learning rate η_t and weight-decay cross term r_t , the coefficient model is

$$F_t(c) = -\eta_t s_t^\top c + \frac{\eta_t^2}{2} (c^\top G_t c + 2r_t^\top c), \quad (5)$$

$$\sum_i \omega_i c_i^2 = \sum_i \omega_i, \quad \omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2.$$

The constraint matches the feed-forward direction norm of $c = 1$ before decay. A Krylov-subspace solve proposes coefficients; acceptance requires $F_t(c) < F_t(1)$, the prescribed norm tolerance, and positive gradient and momentum contractions for every layer and matrix role under both allocations. Otherwise $c = 1$.

The selected allocation gives $A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}}) A_i - \eta_t c_i q_i^A$, and the same rule updates B_i , with decay rate λ_{wd} . Attention matrices use response-guided rotations with layerwise GRAIN summaries and their own gradients and moments. The joint coefficients scale the paired feed-forward directions. Rational coefficients and remaining parameters use AdamW (Loshchilov & Hutter, 2019). Table 1 gives the complete common procedure.

3.2.1 FULL-HORIZON TILLER

Full-horizon TILLER applies the common construction through update T . Response geometry is refreshed periodically, while matrix directions and the loss ledger are updated at every step. The allocation follows the evolving GRAIN response throughout training.

3.2.2 STAGED TRAINING

Staged training applies the same construction through a selected step $T_s < T$, then updates the feed-forward and attention matrices with a chosen optimizer $\mathcal{U}$. Its state initializer $\mathcal{I}$ maps the learned parameters and TILLER state to the continuation moments, counters, and matrix grouping. The transition follows update T_s , so $\mathcal{U}$ first acts at $T_s + 1$. Model parameters and the global learning-rate schedule continue across the boundary. The rational coefficients continue learning with their existing AdamW state. Response collection and ledger updates stop, leaving the subsequent matrix updates and their cost to $\mathcal{U}$.

4 EXPERIMENTS

[TODO: Experiment section]

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with method development, experimental planning, implementation, mathematical derivations and proof checking, literature analysis, result interpretation, and manuscript drafting and revision.

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A GRAIN RESPONSE

The normalization and rescaling in GRAIN let the learned curve operate on relative feature magnitudes while retaining a response to input amplitude. For one group, suppress the layer and group indices and write

$$\Phi(z) = \rho\phi(u), \quad u = z/\rho, \quad \rho = (z^\top z/w + \epsilon_{\text{act}})^{1/2}. \quad (6)$$

The scalar function acts coordinatewise, and its coefficients are held fixed when differentiating with respect to z . Its denominator is positive for finite coefficients. Absolute-value derivatives at zero use $\text{sign}(0) = 0$; at these kinks the Jacobian below is the selected autodifferentiation linearization.

Both the normalized argument and the restored scale contribute to the response. With $f = \phi(u)$, $d_\phi = \phi'(u)$, and $r_\phi = f - u \odot d_\phi$, differentiation gives

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1}(I - uu^\top/w), \quad J = \frac{\partial \Phi}{\partial z} = \text{Diag}(d_\phi) + \frac{r_\phi u^\top}{w}. \quad (7)$$

The diagonal term applies the local rational slopes. The rank-one term couples the coordinates through the shared scale, with r_ϕ measuring the departure from a local linear response.

TILLER needs the action of this response on perturbations and loss sensitivities. These actions follow directly from the rank-one form:

$$Jv = d_\phi \odot v + r_\phi \frac{u^\top v}{w}, \quad J^\top \zeta = d_\phi \odot \zeta + u \frac{r_\phi^\top \zeta}{w}. \quad (8)$$

Each needs one inner product and coordinatewise products after the curve and slope have been evaluated. The incoming path in $B\Phi(Ax)$ consequently acts as $BJUx$ for a matrix direction U , while an outgoing direction V acts as $V\Phi(Ax)$.

The coefficient initialization in the algorithm approximates SiLU on $[-5, 5]$ and is shared only at initialization: all ten coefficients subsequently learn independently in each group. The implemented width is $w = 256$. For a SwiGLU intermediate width H , the GRAIN width is $H_r = Gw = 3H/2$, so its two dense projections contain $2dH_r = 3dH$ weights, matching the three SwiGLU projections.

B RESPONSE-GUIDED MATRIX DIRECTIONS

TILLER chooses a matrix direction before assigning its joint update coefficient. The direction combines two transformations of the same momentum: approximate polar normalization controls its singular values, and factorized adaptation incorporates the history of entrywise gradient scales. The learned activation determines how much of the adaptive component enters; the current gradient determines its orientation. This construction preserves the calibrated parent norm, leaving the coefficient allocation to redistribute that norm across groups.

For a layer-group pair $i = (\ell, g)$ and role $r \in \{A, B\}$, let $g_{t,i}^r$ denote the accumulated, distributed-averaged, clipped gradient. We use $\beta_1 = \beta_2 = 0.95$ and $\epsilon_{\text{opt}} = 10^{-8}$; δ_{32} denotes the smallest positive normal float32 value. From a zero buffer, form

$$v_{t,i}^r = \beta_1 v_{t-1,i}^r + (1 - \beta_1) g_{t,i}^r, \quad m_{t,i}^r = \beta_1 v_{t,i}^r + (1 - \beta_1) g_{t,i}^r. \quad (9)$$

The parent is a groupwise Muon direction (Jordan et al., 2024; Liu et al., 2025),

$$p_{t,i}^r = \chi_{\text{grp}} \mathcal{P}_5(m_{t,i}^r), \quad \chi_{\text{grp}} = \sqrt{\frac{\min\{H_r, d\}}{G \min\{w, d\}}}, \quad \mu_{\text{mlp}} = 0.2 \sqrt{\max\{H_r, d\}}, \quad (10)$$

where $\mathcal{P}_5$ is the five-step Newton-Schulz map in the algorithm and $H_r = Gw$. With exact full-rank polar factors, the sum of squared block norms would be $G \min\{w, d\}$; multiplying by χ_{grp} matches the nominal whole-projection norm $\min\{H_r, d\}$. The multiplier μ_{mlp} is applied after direction selection.

Factorized second moments provide the adaptive transformation (Shazeer & Stern, 2018). Suppressing the matrix indices and initializing both marginals to zero, set

$$R_t = \beta_2 R_{t-1} + (1 - \beta_2) \text{rowsum}(g_t^2), \quad C_t = \beta_2 C_{t-1} + (1 - \beta_2) \text{colsum}(g_t^2), \\ V_{t,jk} = \frac{R_{t,j} C_{t,k}}{\max\{\sum_s R_{t,s}, \delta_{32}\}(1 - \beta_2^2)}, \quad a_t = m_t / (\sqrt{V_t} + \epsilon_{\text{opt}}). \quad (11)$$

The marginals commute with exponential averaging, so they summarize the corresponding row and column sums of the full second-moment history. Their normalized outer product supplies the entrywise scale. Removing a_t 's component parallel to p_t then gives a tangent to the Frobenius sphere through p_t .

The response statistics describe where that tangent should contribute. At a sampled position, incoming perturbations pass through J_{ki} , whereas outgoing perturbations recombine h_{ki} . We quantify their response breadth, for nonzero responses, by

$$\text{pr}_A(J) = \frac{\text{tr}(JJ^\top)^2}{w\|JJ^\top\|_F^2}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w\|h\|_4^4}. \quad (12)$$

The first measures energy spread across squared singular values, and the second across squared feature coordinates. Equal energy in m nonzero components gives m/w . Thus the incoming statistic is spectral, while the outgoing statistic refers to the model's feature coordinates.

Positions are weighted by the loss sensitivity of the corresponding path. For the downstream derivative $\xi_{k\ell}$, define

$$\alpha_{ki}^A = \|J_{ki}^\top B_i^\top \xi_{k\ell}\|_2^2/w, \quad \alpha_{ki}^B = \|\xi_{k\ell}\|_2^2/d, \quad \pi_i^r = \frac{\sum_k \alpha_{ki}^r \text{pr}_r(\mathcal{R}_{ki}^r)}{\sum_k \alpha_{ki}^r}, \quad (13)$$

where $\mathcal{R}_{ki}^A = J_{ki}$ and $\mathcal{R}_{ki}^B = h_{ki}$, and the total sensitivity in each ratio is positive. The incoming weight is the mean squared derivative at the preactivation; the outgoing weight is the corresponding quantity at the block output. This weighting emphasizes response geometry at loss-sensitive positions.

To measure how learning changes the response, evaluate the initial rational coefficients at the same current normalized inputs and scales. For $\Gamma_{ki}^A = J_{ki}J_{ki}^\top$ and $\Gamma_{ki}^B = h_{ki}h_{ki}^\top$, let the superscript 0 denote this reference evaluation. For positive weighted kernel norms, define congruence by

$$\kappa_i^r = \frac{\sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle}{\sqrt{(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2) (\sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2)}}. \quad (14)$$

This is a cosine between collections of response kernels weighted by $\sqrt{\alpha_{ki}^r}$. Evaluating both curves at current inputs isolates the coefficient change.

Set $e_i^r = \pi_i^r(1 - (\kappa_i^r)^2)$. For a nonzero parent and tangent, suppressing time and role indices, the direction before calibration is

$$a_i^\perp = a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, \quad \tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \quad (15)$$

Set $\sigma_i = 1$ when $\langle g_i, a_i^\perp \rangle \geq 0$ and $\sigma_i = -1$ otherwise. The numerical algorithm computes the combination as coefficients of p_i and a_i , and selects it when it is finite, nondegenerate, and has a positive gradient contraction along with the parent. Calibration then gives $q_i = \mu_{\text{mlp}} \tilde{q}_i$; a rejected rotation uses $q_i = \mu_{\text{mlp}} p_i$.

Attention uses the response of the same layer's GRAIN map as a shared signal. It averages participation across groups but aggregates the kernel inner products and squared norms before taking congruence. With these layer statistics, its rotation fraction is

$$e_\ell^{\text{att}} = \sqrt{\pi_\ell^A \pi_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B). \quad (16)$$

The breadth factor includes both feed-forward paths, and departure in either can increase the rotation. QKV and attention-output matrices share this fraction while retaining their own moments, tangents, and gradient-selected signs. Their selected directions are applied directly; joint coefficients are reserved for the paired GRAIN projections.

C JOINT LOSS-RESPONSE COORDINATION

One coefficient per layer-group pair controls the strength of both its incoming and outgoing directions. Directions in disjoint parameter blocks can nevertheless act together through the network. Measuring their contributions at shared activation positions makes those interactions available to the allocation model, while using the full gradient supplies its current descent term.

C.1 SHARED-POSITION MEASUREMENTS

Fix a group $i = (\ell, g)$, a position k , and directions (q_i^A, q_i^B) . Differentiating $(B_i + \alpha q_i^B)\Phi_i((A_i + \alpha q_i^A)x_{k\ell})$ at $\alpha = 0$ gives

$$\begin{aligned}\delta y_{ki} &= B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki}, \\ S_{t,ki} &= \langle q_i^A x_{k\ell}, J_{ki}^\top B_i^\top \xi_{k\ell} \rangle + \langle q_i^B h_{ki}, \xi_{k\ell} \rangle.\end{aligned}\quad (17)$$

The cotangent $\xi_{k\ell}$ is the derivative of the summed minibatch loss at the block output, multiplied by the realized clipping factor. It collects the hidden position’s influence on downstream loss terms. The score consequently measures a contribution through that activation position along the paired direction.

Probe derivatives and parameter gradients must use consistent scaling. For a microbatch with N flattened positions whose mean loss is divided by the accumulation count M , a captured derivative $\bar{\xi}_{k\ell}$ contains the factor $1/(NM)$. TILLER uses

$$\xi_{k\ell} = NM \chi_{\text{clip}} \bar{\xi}_{k\ell}, \quad \chi_{\text{clip}} = \min\{1, (\|g_{\text{raw}}\|_2 + 10^{-6})^{-1}\}.\quad (18)$$

The clipping factor is held fixed while evaluating directions. Averaging the resulting scores over all positions and processes recovers the clipped minibatch derivative under this averaging convention. Its exact value for the current directions is already available from the parameter gradients:

$$s_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle.\quad (19)$$

The linear term uses s_t directly; the K sampled rows of S_t measure how the positional contributions vary together.

Weight decay contributes through the same response paths. Substituting $(\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i)$ for (q_i^A, q_i^B) and summing over groups and layers gives the decay response $a_t^{\text{dec}} \in \mathbb{R}^K$. The combined squared response is

$$\frac{\|S_t c + a_t^{\text{dec}}\|_2^2}{K} = c^\top \frac{S_t^\top S_t}{K} c + 2c^\top \frac{S_t^\top a_t^{\text{dec}}}{K} + \frac{\|a_t^{\text{dec}}\|_2^2}{K}.\quad (20)$$

The last term is independent of the allocation. The first two therefore supply a quadratic response penalty and its decay cross term. Combined with current descent and a fixed direction norm, this penalty compares the recorded joint effects of different allocations at the scale of the applied step.

C.2 DISCOUNTED RESPONSE HISTORY

Responses are measured at steps 1, 9, 17, $\dots$, using $K = 32$ positions aligned across layers. A refresh recomputes participation, congruence, and positional scores at the current parameters. Between refreshes, only the response statistics are reused: the matrix moments, selected directions, and full-gradient contractions are recomputed at every step. Historical columns preserve their layer–group identities, with each row recording the direction in use when it was measured.

The intervening steps update the ledger’s orientation using current descent while retaining the latest measured response scale. Let $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ and let d_t^{dec} be the clipped-gradient contraction with the GRAIN decay direction. The synthetic rows are

$$S_t = \mathbf{1}_K (\gamma_t s_t)^\top, \quad a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}, \quad \gamma_t = \begin{cases} \nu / \|s_t\|_2, & \|s_t\|_2 > 0, \\ 0, & \|s_t\|_2 = 0. \end{cases}\quad (21)$$

For nonzero s_t , their second moment has rank at most one and trace ν^2 . The refreshed rows supply positional variation; the synthetic rows carry current-gradient information between those measurements. The numerical zero threshold is specified in the algorithm.

For the resulting sequence of measured and synthetic rows, the uncompressed target is

$$G_t^* = \beta_2 G_{t-1}^* + (1 - \beta_2) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K.\quad (22)$$

Thus $c^\top G_t^* c$ averages squared responses of a coefficient combination over the recorded history. The decay $\beta_2 = 0.95$ discounts old relationships as the directions and learned responses change.

The ledger represents the previous metric by a row factor and an isotropic remainder, $\hat{G}_{t-1} = \mathcal{B}_{t-1}^\top \mathcal{B}_{t-1} / K + \tau_{t-1} I$. Adding current rows gives the metric used for selection:

$$\begin{aligned} \mathcal{A}_t &= \begin{bmatrix} \sqrt{\beta_2 \mathcal{B}_{t-1}} \\ \sqrt{1 - \beta_2} S_t \end{bmatrix}, & G_t &= \mathcal{A}_t^\top \mathcal{A}_t / K + \beta_2 \tau_{t-1} I, \\ r_t &= \beta_2 r_{t-1} + (1 - \beta_2) S_t^\top a_t^{\text{dec}} / K. \end{aligned} \quad (23)$$

At the first step, $\mathcal{A}_1 = \mathcal{B}_1 = S_1$, $\tau_0 = \tau_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$. The current metric G_t contains the fresh rows before their compression; $\hat{G}_t$ denotes the compressed metric retained for future steps.

Robust Frequent Directions limits the stored factor to $r = 64$ rows (Ghashami et al., 2016; Luo et al., 2019). Let the singular values of $\mathcal{A}_t$ be $\sigma_1 \geq \sigma_2 \geq \dots$. Let r_* be the smaller of r and the number of singular values. When the rank exceeds r , set $\Delta_t = \sigma_{r+1}^2$; otherwise set $\Delta_t = 0$. Store

$$\mathcal{B}_t = \text{Diag}(\sqrt{\sigma_1^2 - \Delta_t}, \dots, \sqrt{\sigma_{r_*}^2 - \Delta_t}) V_{1:r_*}^\top, \quad \tau_t = \beta_2 \tau_{t-1} + \Delta_t / (2K). \quad (24)$$

For rank at most r , this retains an exact factor. The numerical algorithm obtains the factor from the row Gram matrix. Its dominant rows retain response combinations, while the isotropic term centers the approximation error across coefficient directions.

C.3 FIXED-NORM ALLOCATION

The allocation compares different strengths at the norm of the current unit-allocation update. With $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$ and $\mathcal{E}_0 = \sum_i \omega_i$, disjoint parameter blocks give $\|\Delta\theta(c)\|_2^2 = \eta_t^2 \sum_i \omega_i c_i^2$. Consequently, the constraint $\sum_i \omega_i c_i^2 = \mathcal{E}_0$ matches the feed-forward direction norm of $c = \mathbf{1}$ before weight decay. Signed coefficients let the model compare the combined actions of groups and layers while preserving the relative calibration of each group’s incoming and outgoing paths.

Whitening the constraint turns the proposal calculation into a spherical quadratic problem. For positive ω_i , set $W = \text{Diag}(\omega)$ and $y = W^{1/2} c$. Dividing the coefficient model by η_t^2 gives

$$\begin{aligned} Q_t(y) &= \frac{1}{2} y^\top M_t y - b_t^\top y, & \|y\|_2^2 &= \mathcal{E}_0, \\ M_t &= W^{-1/2} G_t W^{-1/2}, & b_t &= W^{-1/2} (s_t / \eta_t - r_t). \end{aligned} \quad (25)$$

Its stationarity equation is $(M_t + \lambda I)y = b_t$. The positive-multiplier branch gives a positive-definite system because M_t is positive semidefinite. TILLER uses this branch to construct a proposal in a subspace of at most $\min\{32, C\}$ dimensions. The proposal is defined for positive group energies, positive learning rate, and nonzero b_t .

The subspace is built from repeated metric actions starting at b_t using two reorthogonalization passes and orthogonal restarts (Lanczos, 1950). The factor representation gives

$$M_t v = W^{-1/2} \mathcal{A}_t^\top (\mathcal{A}_t W^{-1/2} v) / K + \beta_2 \tau_{t-1} W^{-1} v. \quad (26)$$

Let Q_k be the resulting basis and T_k the accumulated symmetric tridiagonal matrix. Clipping its negative eigenvalues gives $T_k^+ = U \text{Diag}(\theta) U^\top$ with $\theta_j \geq 0$. For $\gamma = U^\top Q_k^\top b_t$, the projected proposal is

$$y = Q_k U \text{Diag}((\theta + \lambda)^{-1}) \gamma, \quad \psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (27)$$

The prescribed bracket and 64 bisection steps produce $c = W^{-1/2} y$ when a positive root is bracketed. If the lower-end test rules out a root in this bracket, TILLER retains the unit allocation.

The projected model proposes coefficients; the original factor metric decides whether to use them. Acceptance requires a finite candidate, $F_t(c) < F_t(\mathbf{1})$, and scaled norm residual at most 10^{-8} , measured in float64. For every layer and role, it also requires

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad r \in \{A, B\}. \quad (28)$$

The same inequalities must hold for $c = \mathbf{1}$. Rejected proposals retain the unit allocation; accepted coefficients are cast to float32 before application to the matrix directions.

D PROPERTIES OF THE CONSTRUCTION

D.1 ACTIVATION AND UPDATE GEOMETRY

The GRAIN Jacobian separates changes in relative features from changes in group amplitude. A direction v with $u^\top v = 0$ leaves the scale unchanged to first order and satisfies $Jv = d_\phi \odot v$. Along the input, $u^\top u/w = 1 - \epsilon_{\text{act}}/\rho^2$ yields

$$Jz = \Phi(z) - \frac{\epsilon_{\text{act}}}{\rho} r_\phi. \quad (29)$$

For $\epsilon_{\text{act}} = 0$ and nonzero z , positive rescaling obeys $\Phi(az) = a\Phi(z)$ for $a > 0$, and the identity becomes $Jz = \Phi(z)$. The correction term quantifies the stabilizer’s effect on the radial response. For a linear curve $\phi(u) = au$, $r_\phi = 0$ and $\Phi(z) = az$, so the scale operations cancel even with a positive stabilizer.

Participation and congruence define a bounded rotation fraction. For the ratios in equation 12, nonnegative component energies v_j satisfy $\sum_j v_j^2 \leq (\sum_j v_j)^2 \leq w \sum_j v_j^2$, placing nonzero participation in $[1/w, 1]$. The identities $\langle JJ^\top, J^0 J^{0\top} \rangle = \|J^\top J^0\|_F^2$ and $\langle hh^\top, h^0 h^{0\top} \rangle = (h^\top h^0)^2$, together with Cauchy–Schwarz, place defined congruence in $[0, 1]$. The zero-response conventions in the algorithm keep e_i^r in this interval and retain the parent when reference alignment is undefined.

Congruence measures alignment of response energy. Sign reversals of J or h preserve their kernels, and a common nonzero gain $\phi = c\phi^0$ gives $\kappa = 1$. The loss-response measurements retain the gain through J and h .

Orthogonality gives $\|\tilde{q}_i\|_F^2 = (1 - e_i)\|p_i\|_F^2 + e_i\|p_i\|_F^2$. Choosing σ_i by the sign of $\langle g_i, a_i^\perp \rangle$ makes the tangent contribute nonnegatively to the descent contraction for a subtraction update. The calibrated selection has norm $\|\mu_{\text{mlp}} p_i\|_F$ in exact arithmetic when numerical floors are inactive; parent fallback has that norm directly.

Participation attenuates the angular departure from the reference. With $e_i^r = \pi_i^r(1 - (\kappa_i^r)^2)$, the rotation angle ϑ_i^r and kernel angle $\psi_i^r = \arccos \kappa_i^r$ obey $\sin^2 \vartheta_i^r = \pi_i^r \sin^2 \psi_i^r$. A concentrated response therefore permits a smaller adaptive component, while full participation permits $\vartheta_i^r = \psi_i^r$.

The coefficient proposal enforces its prescribed norm through the scalar equation $\psi(\lambda) = \mathcal{E}_0$. For nonzero γ and $\lambda > 0$,

$$\psi'(\lambda) = -2 \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^3} < 0. \quad (30)$$

Thus the positive branch has at most one root. For an orthonormal basis and nonnegative projected eigenvalues, the algorithm’s upper endpoint satisfies $\psi(\lambda_{\text{hi}}) \leq \|b_t\|_2^2/\lambda_{\text{hi}}^2 \leq \mathcal{E}_0$. The lower-end test establishes whether this bracket contains a root.

D.2 RESPONSE-HISTORY APPROXIMATION

The retained factor approximates a specified discounted response history. Its error can be bounded independently of coefficient selection because Robust Frequent Directions applies the same singular-value shrinkage to each retained direction and stores half that amount isotropically. The following result concerns this representation in exact arithmetic.

Proposition 1 (Response-metric approximation). *For $K > 0$, $\beta_2 \in [0, 1]$, and $\tau_{t-1} \geq 0$, let G_t and $(\mathcal{B}_t, \tau_t)$ be defined by equations 23 and 24. Then $\hat{G}_t = \mathcal{B}_t^\top \mathcal{B}_t/K + \tau_t I$ is positive semidefinite, and for every $c \in \mathbb{R}^C$,*

$$\|\hat{G}_t - G_t\|_2 \leq \frac{\Delta_t}{2K}, \quad |c^\top (\hat{G}_t - G_t) c| \leq \frac{\Delta_t}{2K} \|c\|_2^2. \quad (31)$$

Proof. Both terms of $\hat{G}_t$ are positive semidefinite. For $\Delta_t = 0$, the two metrics coincide. For positive Δ_t , work in a right-singular-vector basis of $\mathcal{A}_t$, completed by its null space. Each retained squared singular value loses Δ_t/K and receives $\Delta_t/(2K)$ from the isotropic term, giving error $-\Delta_t/(2K)$. A discarded direction with squared singular value $s \in [0, \Delta_t]$ has error $(\Delta_t/2 - s)/K$;

a null direction has error $\Delta_t/(2K)$. Every absolute eigenvalue of the difference is therefore at most $\Delta_t/(2K)$. The quadratic-form bound follows from $|c^\top Ec| \leq \|E\|_2 \|c\|_2^2$ for symmetric E . $\square$

The isotropic increment centers the one-step compression error. The same retained factor with the previous remainder $\beta_2 \tau_{t-1} I$ has error eigenvalues in $[-\Delta_t/K, 0]$. Adding $\Delta_t I/(2K)$ centers this interval at zero and halves the one-compression spectral bound, from Δ_t/K to $\Delta_t/(2K)$, for the response metric retained for subsequent steps.

Discounting limits how older compression errors accumulate. Write $E_t = \hat{G}_t - G_t^*$ and $D_t = \hat{G}_t - G_t$. The common initialization gives $E_1 = 0$, and the two recurrences imply $E_t = \beta_2 E_{t-1} + D_t$. Hence

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta_2^{t-j} \frac{\Delta_j}{2K}, \quad G_t - G_t^* = \beta_2 E_{t-1}. \quad (32)$$

The current metric carries only previous compression error because it includes fresh rows before the new shrinkage. These bounds describe fidelity to the measured-and-synthetic response history and its quadratic penalty.

E COMPLETE TRAINING ALGORITHM

Full-horizon and staged training use the same TILLER update. A horizon T and transition index $T_s \in \{1, \dots, T\}$ specify how long it is applied. When $T_s < T$, a continuation rule $\mathcal{U}$ and its state initializer $\mathcal{I}$ complete the procedure. The initializer maps the parameters and optimizer state after update T_s to the moments, counters, and matrix grouping needed by $\mathcal{U}$. The model and global learning-rate schedule continue across this boundary, as does the AdamW state for rational coefficients and the remaining parameter classes.

Rows 1–7 of Table 1 give the training loop; the remaining rows define its local operations.

Table 1: GRAIN training with full-horizon or staged TILLER. All local operations used by the training loop are specified in this table. Setting $T_s = T$ gives full-horizon TILLER; otherwise $(\mathcal{U}, \mathcal{I})$ defines the continuation.

Procedure and local operations
Inputs: initial model parameters, L layers, G groups of width $w = 256$, model width d , $H_r = Gw$, $C = LG \geq 2$, horizon T , transition $T_s \in \{1, \dots, T\}$, schedule η_t , matrix decay λ_{wd} , a model and loss averaged over the microbatch's N token positions, a minibatch iterator, $M = 4$ accumulation steps, P distributed processes, and per-parameter AdamW schedules $\eta_\theta(t)$ and decay rates. The grouped matrices have $A_i \in \mathbb{R}^{w \times d}$ and $B_i \in \mathbb{R}^{d \times w}$, $i = (\ell, g)$. For $T_s < T$, supply $\mathcal{U}$ and $\mathcal{I}$. Require $\eta_t > 0$ during TILLER. The supported attention layout has stacked QKV of shape $3d \times d$ and an output matrix of shape $d \times d$, with 16 heads, four heads per polar block, and $16 \mid d$. Constants: $\beta_1 = \beta_2 = 0.95$, $\epsilon_{act} = 10^{-6}$, $\epsilon_{opt} = 10^{-8}$, $K = 32$, refresh interval 8, persistent row limit 64, Krylov width $\min\{32, C\}$, and 64 bisection steps. The matrix decay default is $\lambda_{wd} = 0.1$. Rational coefficients, tied embeddings, normalization parameters, and biases have zero AdamW decay; other AdamW matrices use 0.1. Write δ_{32} and δ_{64} for the smallest positive normal float32 and float64 values, and ϵ_{64} for float64 machine epsilon. Inner products of matrices are Frobenius inner products. Elementwise powers, products, and division are used for moment updates.

Table 1 (continued)

Procedure and local operations
For every group, initialize the numerator and denominator coefficients, in ascending powers, to
$a^0 = (2.4915714620520875 \times 10^{-5}, 0.5000668663485823, \\ 0.2499441908995771, 0.0526200910151016, \\ 0.005525765640115002, 0.00024199321178747251),$ $b^0 = (0.00022554017910995938, 0.10511420350691994, \\ 2.8656992394109812 \times 10^{-5}, 0.0004816962110562025).$
During the forward pass, each group computes $z_g = A_g x$, $\rho_g = \sqrt{\ z_g\ ^2/w + \epsilon_{\text{act}}}$, $h_g = \rho_g P_g(z_g/\rho_g)/D_g(z_g/\rho_g)$ with the polynomials defined in row 9, and the block returns $\sum_g B_g h_g$. Keep a frozen copy (a^0, b^0) for response comparisons. Initialize each matrix's momentum v and row/column second moments R, C_{col} to zero, initialize AdamW moments and counts to zero, and leave the response ledger and response cache empty.
- For $t = 1, \dots, T$, clear parameter gradients and let the refresh flag be $t \leq T_s$ and $(t-1) \bmod 8 = 0$. At the current parameters, evaluate each microbatch mean loss, divide by the accumulation count M, and backpropagate. On a refresh, capture the probe positions and their block inputs, preactivations, features, and block-output derivatives using the selection rule below. Distributed-average the accumulated parameter gradients. Evaluate all responses and directions before applying any parameter update. - With g_{raw} the concatenated gradient, set $\chi = \min\{1, (\ g_{\text{raw}}\ _2 + 10^{-6})^{-1}\}$ and $g = \chi g_{\text{raw}}$. Nonfinite gradients are errors. For a captured microbatch with N flattened positions and captured derivative $\bar{\xi}$, set $\xi = NM\chi\bar{\xi}$. - If $t \leq T_s$, refresh and cache the response summaries in rows 8–12 when the refresh flag is set. Construct all feed-forward and attention directions using rows 13–15. - When $t \leq T_s$, for each $i = (\ell, g)$ and $r \in \{A, B\}$, form $d_i^r = \langle g_i^r, q_i^r \rangle$, $u_i^r = \langle m_i^r, q_i^r \rangle$, $s_i = d_i^A + d_i^B$ (summing the two float32 contractions in float64), and $\omega_i = \ q_i^A\ _{\mathbb{F}}^2 + \ q_i^B\ _{\mathbb{F}}^2$. Construct current response rows and the ledger factor using rows 16–19. - Still within $t \leq T_s$, apply rows 20–23 to the current factor, isotropic term, decay cross term, (d_i^r, u_i^r), ω, and η_t, obtaining c. Retain (B, τ, r) for the next step. Apply
$A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}})A_i - \eta_t c_i q_i^A, \quad B_i \leftarrow (1 - \eta_t \lambda_{\text{wd}})B_i - \eta_t c_i q_i^B.$
For each attention matrix W , apply $W \leftarrow (1 - \eta_t \lambda_{\text{wd}})W - \eta_t q_W$. Directions and selected coefficients are float32. Retain the unscaled selected direction $\tilde{q}$ and its matrix-size multiplier μ from row 15. For application, cast $c_i \tilde{q}_i^r$ (or $\tilde{q}_W$ for attention) to the parameter dtype, then add it with scalar multiplier $-\eta_t \mu$.
- If $t > T_s$, update the feed-forward and attention matrices with the supplied rule $\mathcal{U}$, its state, and the current clipped gradients at learning rate η_t. In either branch, update rational coefficients and the remaining parameters by the AdamW operation below. - After completing update $t = T_s < T$, set the continuation state to $\mathcal{I}$(current parameters, current optimizer state), retaining the learned parameters and AdamW state. Stop response collection and ledger updates. Continue the loop with the same global schedule. After $t = T$, return the trained model. Rows 8–23 and the AdamW operation define the local operations invoked by rows 1–7.
<i>Probe selection and response summaries.</i>
For P processes, process $p \in \{0, \dots, P-1\}$ contributes $K_p = \lfloor (p+1)K/P \rfloor - \lfloor pK/P \rfloor$ positions. Require each selection to satisfy $0 \leq m \leq N$. From each of the M microbatches capture $m_p = \lceil K_p/M \rceil$ positions, then select K_p from their concatenation. Selecting $m > 1$ rows from N rows uses zero-based indices $\lfloor j(N-1)/(m-1) \rfloor$, $j = 0, \dots, m-1$; $m = 1$ selects zero and $m = 0$ selects nothing. Use identical indices across layers and retain the captured tensors locally.

Table 1 (continued)

Procedure and local operations

- 9 Cast captured records and rational coefficients to float32; evaluate the response and statistics in rows 9–12 in float32. At each selected position and group, set $\rho = \sqrt{z^\top z}/w + \epsilon_{\text{act}}$ and $u = z/\rho$. For current or frozen coefficients evaluate

$$P(u) = \sum_{j=0}^5 a_j u^j, \quad D(u) = 1 + \sum_{j=1}^4 |b_j| |u|^j,$$

$$f = P(u)/D(u), \quad d_\phi = \frac{P'(u)D(u) - P(u)D'(u)}{D(u)^2}, \quad r_\phi = f - u \odot d_\phi,$$

$$P'(u) = \sum_{j=1}^5 j a_j u^{j-1}, \quad D'(u) = \sum_{j=1}^4 j |b_j| |u|^{j-1} \text{sign}(u).$$

- Use $\text{sign}(0) = 0$. For response statistics set $h^{\text{eval}} = \rho f$ and $J = \text{Diag}(d_\phi) + r_\phi u^\top/w$. Apply $Jv = d_\phi \odot v + r_\phi(u^\top v)/w$ and $J^\top v = d_\phi \odot v + u(r_\phi^\top v)/w$ without storing J . Frozen evaluation uses the same current u and ρ . Retain the captured forward feature h separately for row 16.
- 10 Define $\Gamma^A = JJ^\top$ and $\Gamma^B = h^{\text{eval}}(h^{\text{eval}})^\top$ through their inner products. The diagonal-plus-rank-one Jacobian makes $JJ^\top$ a diagonal matrix plus a rank-two correction, so these inner products reduce to vector contractions and a 2×2 matrix calculation. For two Jacobians at the same u , let $r = r_\phi/w$, $\tilde{r} = \tilde{r}_\phi/w$, $v = d_\phi \odot u$, $\tilde{v} = \tilde{d}_\phi \odot u$, $s = u^\top u$, $Z = [v, r]^\top [\tilde{v}, \tilde{r}]$, and $E = \begin{bmatrix} 0 & 1 \\ 1 & s \end{bmatrix}$. Then evaluate

$$\begin{aligned} \langle \Gamma^A, \tilde{\Gamma}^A \rangle &= \langle d_\phi^2, \tilde{d}_\phi^2 \rangle + 2\langle d_\phi^2, \tilde{v} \odot \tilde{r} \rangle + s\langle d_\phi^2, \tilde{r}^2 \rangle \\ &\quad + 2\langle \tilde{d}_\phi^2, v \odot r \rangle + s\langle \tilde{d}_\phi^2, r^2 \rangle + \langle EZE, Z \rangle, \\ \text{tr}(JJ^\top) &= \|d_\phi\|^2 + 2\langle d_\phi \odot u, r \rangle + s\|r\|^2, \\ \langle \Gamma^B, \tilde{\Gamma}^B \rangle &= ((h^{\text{eval}})^\top \tilde{h}^{\text{eval}})^2. \end{aligned}$$

Self-inner products give the squared kernel norms.

- 11 For each role compute per-position participation and sensitivity

$$P_{ki}^A = \frac{\text{tr}(J_{ki} J_{ki}^\top)^2}{w \max\{\|J_{ki} J_{ki}^\top\|_F^2, \delta_{32}\}}, \quad P_{ki}^B = \frac{\|f_{ki}\|_2^4}{w \max\{\|f_{ki}\|_4^4, \delta_{32}\}},$$

$$\alpha_{ki}^A = \|J_{ki}^\top B_i^\top \xi_{k\ell}\|^2/w, \quad \alpha_{ki}^B = \|\xi_{k\ell}\|^2/d.$$

Clip only the participation ratios to $[0, 1]$; a zero response gives zero. Here $f = h^{\text{eval}}/\rho$, which gives the same outgoing participation as h^{eval} when the floor is inactive. Accumulate $D_i^r = \sum_k \alpha_{ki}^r$, $N_i^r = \sum_k \alpha_{ki}^r P_{ki}^r$, $X_i^r = \sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle$, $Y_i^r = \sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2$, $Z_i^r = \sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2$. Sum these statistics across processes.

- 12 Set $\pi_i^r = \text{clip}_{[0,1]}(N_i^r / \max(D_i^r, \delta_{32}))$. If either role has $D_i^r = 0$, set both π_i^r to one. Set $\kappa_i^r = \text{clip}_{[0,1]}(X_i^r / \max(\sqrt{Y_i^r Z_i^r}, \delta_{32}))$. If either unfloored denominator $\sqrt{Y_i^r Z_i^r}$ is zero, or the group's coefficients equal their initialization, set both congruences to one. For layer ℓ , set $\bar{\pi}_\ell^r = G^{-1} \sum_g \pi_{\ell g}^r$ and $\kappa_\ell^r = (\sum_g X_{\ell g}^r) / \max\{\sqrt{(\sum_g Y_{\ell g}^r)(\sum_g Z_{\ell g}^r)}, \delta_{32}\}$, clipped to $[0, 1]$. A

zero unfloored denominator $\sqrt{(\sum_g Y_{\ell g}^r)(\sum_g Z_{\ell g}^r)}$ for either role sets both layer congruences to one; so does equality to initialization for every group. Cache

$$e_i^r = \pi_i^r (1 - (\kappa_i^r)^2), \quad e_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B).$$

All response statistics must be finite.

Table 1 (continued)

 Procedure and local operations

Matrix moments and direction construction.

- 13 For each TILLER-assigned matrix, update its first moment in the gradient dtype:
 $v \leftarrow \beta_1 v + (1 - \beta_1)g$ and $m = \beta_1 v + (1 - \beta_1)g$. For each feed-forward group block, and for each whole attention matrix, update

$$R \leftarrow \beta_2 R + (1 - \beta_2) \text{rowsum}(g^2), \quad C_{\text{col}} \leftarrow \beta_2 C_{\text{col}} + (1 - \beta_2) \text{colsum}(g^2),$$

$$V = \frac{RC_{\text{col}}^\top}{\max\{\mathbf{1}^\top R, \delta_{32}\}(1 - \beta_2^t)}, \quad a = m/(\sqrt{V} + \epsilon_{\text{opt}}).$$

Second moments and the direction arithmetic use float32.

- 14 To compute $\mathcal{P}_5(M)$, transpose a tall block M , cast to bfloat16, compute its Frobenius norm in float32, floor the norm at 10^{-7} , cast the norm to bfloat16, and divide the block by it. Starting with the resulting X , perform five bfloat16 iterations

$$H = XX^\top, \quad X \leftarrow 3.4445X + (-4.775H + 2.0315H^2)X.$$

Restore the original orientation. For each GRAIN matrix block set $p = \chi_{\text{grp}} \mathcal{P}_5(m)$ with $\chi_{\text{grp}} = \sqrt{\min\{H_r, d\} / (G \min\{w, d\})}$. For QKV, apply $\mathcal{P}_5$ to four contiguous row blocks of width $d/4$ in each of Q, K, and V, reassemble, and multiply by $1/\sqrt{3}$. For the attention output matrix, apply it to four contiguous column blocks of width $d/4$ and reassemble. Apply these parent calibration factors in bfloat16, then cast the parents to float32.

- 15 Use $e = e_i^r$ for feed-forward block (i, r) and $e = e_\ell^{\text{att}}$ for both attention matrices in layer ℓ . For the corresponding parent p , adaptive direction a , current gradient g , and cached $e \in [0, 1]$, compute

$$P = \|p\|_F^2, \quad A = \|a\|_F^2, \quad h = \langle p, a \rangle, \quad \zeta = h / \max(P, \delta_{32}),$$

$$D_\perp = \max\{A - h^2 / \max(P, \delta_{32}), 0\}, \quad \sigma = \begin{cases} 1, & \langle g, a \rangle - \zeta \langle g, p \rangle \geq 0, \\ -1, & \text{otherwise,} \end{cases}$$

$$c_a = \sigma \sqrt{e} \sqrt{P / \max(D_\perp, \delta_{32})}, \quad c_p = \sqrt{1 - e} - c_a \zeta, \quad D_{\text{cand}} = c_p \langle g, p \rangle + c_a \langle g, a \rangle.$$

Use (c_p, c_a) only if $p, m, g, a, c_p, c_a, D_{\text{cand}}$ are finite, $P, A > 0$, $D_\perp > \delta_{32}$, $\langle g, p \rangle > 0$, and $D_{\text{cand}} > 0$; otherwise use $(1, 0)$. The selected direction $\tilde{q} = c_p p + c_a a$ must be finite. Set $\mu = 0.2 \sqrt{\max\{H_r, d\}}$ for GRAIN blocks and $\mu = 0.2 \sqrt{\max\{n, m\}}$ for a whole $n \times m$ attention matrix. The calibrated direction is $q = \mu \tilde{q}$.

Response rows and temporal ledger.

- 16 On refresh steps, for every locally selected position and group form

$$S_{ki} = \langle q_i^A x_{k\ell}, J_{ki}^\top B_i^\top \xi_{k\ell} \rangle + \langle q_i^B h_{ki}, \xi_{k\ell} \rangle,$$

$$a_k^{\text{dec}} = \lambda_{\text{wd}} \sum_i (\langle A_i x_{k\ell}, J_{ki}^\top B_i^\top \xi_{k\ell} \rangle + \langle B_i h_{ki}, \xi_{k\ell} \rangle).$$

Gather S and a^{dec} across processes to obtain the K aligned rows. Save $\nu = \|S\|_F / \sqrt{K}$, accumulating its squared norm in float64 and storing the result in float32.

- 17 Between refreshes compute in float32 $d^{\text{dec}} = \lambda_{\text{wd}} \sum_i (\langle g_i^A, A_i \rangle + \langle g_i^B, B_i \rangle)$. In float64 compute $\gamma = \nu / \|s\|_2$ if $\|s\|_2 > \delta_{64}$ and $\gamma = 0$ otherwise. Cast γs and γd^{dec} to float32 and repeat them in all K rows: $S = \mathbf{1}_K (\gamma s)^\top$, $a^{\text{dec}} = \mathbf{1}_K \gamma d^{\text{dec}}$. Require finite rows, contractions, and $\nu \geq 0$.
- 18 At the first step set $\mathcal{A} = S$, $\delta = 0$, $r = S^\top a^{\text{dec}} / K$, and save $(\mathcal{B}, \tau) = (S, 0)$. On later steps, with previous ledger $(\mathcal{B}_-, \tau_-, r_-)$, form

$$\mathcal{A} = \begin{bmatrix} \sqrt{\beta_2} \mathcal{B}_- \\ \sqrt{1 - \beta_2} S \end{bmatrix}, \quad \delta = \beta_2 \tau_-, \quad r = \beta_2 r_- + (1 - \beta_2) S^\top a^{\text{dec}} / K.$$

The current metric is $G = \mathcal{A}^\top \mathcal{A} / K + \delta I$. Keep $\mathcal{A}$ in float32; accumulate the decay cross term in float64 and store it in float32. Require finite factors and cross terms and nonnegative δ . Keep δ in float64 for row 19 and pass a float32 copy, together with the stored r , to the current coefficient operation.

Table 1 (continued)

	Procedure and local operations
19	After the first step, form the next-step factor from the float64 symmetric Gram matrix $H = (\mathcal{A}\mathcal{A}^\top + (\mathcal{A}\mathcal{A}^\top)^\top)/2$ and diagonalize it as $H = U \text{Diag}(\lambda)U^\top$, with eigenvalues sorted in increasing order and clipped below at zero. Let n be its dimension, $j = \min\{64, n\}$, and $\Delta = \lambda_{n-j}$ when $n > j$, otherwise $\Delta = 0$ (indices start at one). For the largest j eigenvalues λ_+ and corresponding columns U_+, save $\mathcal{B} = \text{Diag}\left(\sqrt{\frac{\max(\lambda_+ - \Delta, 0)}{\max(\lambda_+, \delta_{64})}}\right) U_+^\top \mathcal{A}, \quad \tau = \delta + \Delta/(2K).$ Compute the diagonal $\text{diag}(\mathcal{B}^\top \mathcal{B}/K) + \tau \mathbf{1}$ in float64, then cast it and $\mathcal{B}, \tau$ to float32 for storage. Require finite output and $\tau, \Delta \geq 0$. The first step retains $(S, 0)$ directly.
	<i>Projected coefficient proposal and acceptance.</i>
20	Require finite $\mathcal{A}, \delta, r, d_i^r, u_i^r, \omega$, with $\delta \geq 0$, every $\omega_i > 0$, and $\eta > 0$. Convert inputs to float64. Set $d_\omega = \sqrt{\omega}$, $E_0 = \sum_i \omega_i$, $b = (s/\eta - r) \odot d_\omega$, and $\mathcal{Z}_{:i} = \mathcal{A}_{:i}/(\sqrt{K} d_{\omega,i})$, where $\odot$ denotes elementwise division. Require finite $\ b\ _2 > 0$. Invalid inputs are errors. Use the matrix action $\mathcal{M}(v) = \mathcal{Z}^\top (\mathcal{Z}v) + (\delta \odot \omega) \odot v.$
21	Let $k = \min\{32, C\}$, $v_1 = b/\max(\ b\ _2, \delta_{64})$, $v_0 = 0$, $\beta_0 = 0$, and Q_0 have no columns. For $j = 1, \dots, k$, set $Q_j = [Q_{j-1}, v_j]$, compute $a_j = \mathcal{M}(v_j)$, $\alpha_j = v_j^\top a_j$, and $z = a_j - \alpha_j v_j - \beta_{j-1} v_{j-1}$. Perform $z \leftarrow z - Q_j(Q_j^\top z)$ twice and set $\hat{\beta}_j = \ z\ _2$. If $j < k$, set $\epsilon_j = 64\epsilon_{64} \max\{\ a_j\ _2, 1\}$ and start with $v_{j+1} = z/\max(\hat{\beta}_j, \delta_{64})$. For $j < k$, when $\hat{\beta}_j \leq \epsilon_j$, take the seed $f_a = \cos(a_j)$ for $a = 1, \dots, C$, perform $f \leftarrow f - Q_j(Q_j^\top f)$ twice, and replace v_{j+1} by $f/\ f\ _2$ if $\ f\ _2 > \epsilon_j$. For $j < k$, set $\beta_j = 0$ when $\hat{\beta}_j \leq \epsilon_j$, and $\beta_j = \hat{\beta}_j$ otherwise.
22	Form symmetric tridiagonal T_k with diagonal α_j and off-diagonal β_j, and $Q = Q_k$. Require finite Q, T_k and $\ Q^\top Q - I\ _F \leq 2 \times 10^{-10} k$; a failed basis check is an error. Diagonalize $T_k = U \text{Diag}(\theta)U^\top$, replace θ by $\max(\theta, 0)$, and set $\gamma = U^\top Q^\top b$. Define $\psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2}, \quad \lambda_{\text{lo}} = \epsilon_{64} C, \quad \lambda_{\text{hi}} = \ b\ _2/\sqrt{E_0} + \max_j \theta_j + \lambda_{\text{lo}}.$
	Record whether $\psi(\lambda_{\text{lo}}) < E_0$.
23	Repeat 64 times: set $\lambda = (\lambda_{\text{lo}} + \lambda_{\text{hi}})/2$; if $\psi(\lambda) > E_0$, replace λ_{lo} by λ; otherwise replace λ_{hi} by λ. Construct $c_* = [QU(\gamma/(\theta + \lambda_{\text{hi}}))] \odot d_\omega$. Evaluate both c_* and $\mathbf{1}$ in $F(c) = -\eta \sum_i s_i c_i + \frac{\eta^2}{2} \left(\ \mathcal{A}c\ _2^2/K + \delta \ c\ _2^2 + 2r^\top c \right).$
	Accept c_* only if the recorded lower-end condition was false, the candidate and objective are finite, $F(c_*) < F(\mathbf{1})$, and $\sum_i \omega_i c_{*,i}^2 - E_0 /\max\{E_0, 1\} \leq 10^{-8}$. For every layer ℓ and role r, also require $\sum_g c_{*,\ell g} d_{\ell g}^r > 0$, $\sum_g c_{*,\ell g} u_{\ell g}^r > 0$, $\sum_g d_{\ell g}^r > 0$, and $\sum_g u_{\ell g}^r > 0$. Return the accepted coefficient vector cast to float32; otherwise return $\mathbf{1}$.
	<i>AdamW operation.</i> For each assigned parameter θ with gradient g_θ, current scheduled learning rate $\eta_\theta(t)$, and decay λ_θ, advance its count n and moments $m_\theta \leftarrow 0.9m_\theta + 0.1g_\theta$, $v_\theta \leftarrow 0.95v_\theta + 0.05g_\theta^2$. Apply $\theta \leftarrow (1 - \eta_\theta(t)\lambda_\theta)\theta - \eta_\theta(t) \frac{m_\theta/(1 - 0.9^n)}{\sqrt{v_\theta/(1 - 0.95^n)} + 10^{-8}}.$
	These moments and counts continue across the staged transition.